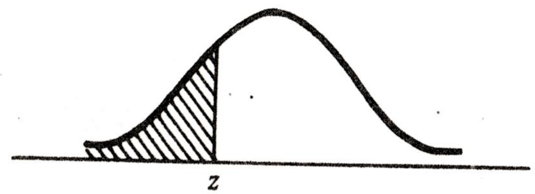
TABLA D Valores de la función de distribución acumulativa normal estándar

$$P(Z \leq z) = F(z; 0, 1) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} \exp(-t^2/2)dt$$



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183

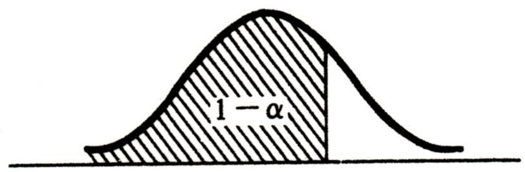
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2297	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7703	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389

TABLA D (continuación) Valores de la función de distribución acumulativa normal estándar

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998

TABLA F Valores de cuantiles de la distribución t de Student

$$P(T \leq t_{1-\alpha,\nu}) = \frac{\Gamma[(\nu+1)/2]}{\sqrt{\pi\nu}\,\Gamma(\nu/2)} \int_{-\infty}^{t_{1-\alpha,\nu}} [1 + (t^2/\nu)]^{-(\nu+1)/2} dt = 1 - \alpha$$



$t_{1-\alpha,\nu}$

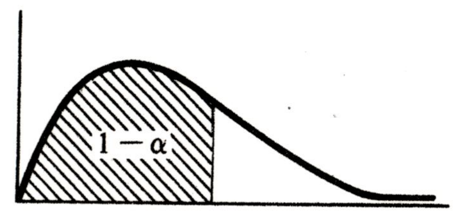
ν	$t_{0.001}$	$t_{0.005}$	$t_{0.010}$	$t_{0.025}$	$t_{0.050}$	$t_{0.100}$	$t_{0.200}$
1	−318.309	−63.657	−31.821	−12.706	−6.314	−3.078	−1.376
2	−22.327	−9.925	−6.965	−4.303	−2.920	−1.886	−1.061
3	−10.215	−5.841	−4.541	−3.182	−2.353	−1.638	−0.978
4	−7.173	−4.604	−3.747	−2.776	−2.132	−1.533	−0.941
5	−5.893	−4.032	−3.365	−2.571	−2.015	−1.476	−0.920
6	−5.208	−3.707	−3.143	−2.447	−1.943	−1.440	−0.906
7	−4.785	−3.499	−2.998	−2.365	−1.895	−1.415	−0.896
8	−4.501	−3.355	−2.896	−2.306	−1.860	−1.397	−0.889
9	−4.297	−3.250	−2.821	−2.262	−1.833	−1.383	−0.883
10	−4.144	−3.169	−2.764	−2.228	−1.812	−1.372	−0.879
11	−4.025	−3.106	−2.718	−2.201	−1.796	−1.363	−0.876
12	−3.930	−3.055	−2.681	−2.179	−1.782	−1.356	−0.873
13	−3.852	−3.012	−2.650	−2.160	−1.771	−1.350	−0.870
14	−3.787	−2.977	−2.624	−2.145	−1.761	−1.345	−0.868
15	−3.733	−2.947	−2.602	−2.131	−1.753	−1.341	−0.866
16	−3.686	−2.921	−2.583	−2.120	−1.746	−1.337	−0.865
17	−3.646	−2.898	−2.567	−2.110	−1.740	−1.333	−0.863
18	−3.610	−2.878	−2.552	−2.101	−1.734	−1.330	−0.862
19	−3.579	−2.861	−2.539	−2.093	−1.729	−1.328	−0.861
20	−3.552	−2.845	−2.528	−2.086	−1.725	−1.325	−0.860
21	−3.527	−2.831	−2.518	−2.080	−1.721	−1.323	−0.859
22	−3.505	−2.819	−2.508	−2.074	−1.717	−1.321	−0.858
23	−3.485	−2.807	−2.500	−2.069	−1.714	−1.319	−0.858
24	−3.467	−2.797	−2.492	−2.064	−1.711	−1.318	−0.857
25	−3.450	−2.787	−2.485	−2.060	−1.708	−1.316	−0.856
26	−3.435	−2.779	−2.479	−2.056	−1.706	−1.315	−0.856
27	−3.421	−2.771	−2.473	−2.052	−1.703	−1.314	−0.855
28	−3.408	−2.763	−2.467	−2.048	−1.701	−1.313	−0.855
29	−3.396	−2.756	−2.462	−2.045	−1.699	−1.311	−0.854
30	−3.385	−2.750	−2.457	−2.042	−1.697	−1.310	−0.854
35	−3.340	−2.724	−2.438	−2.030	−1.690	−1.306	−0.852
40	−3.307	−2.704	−2.423	−2.021	−1.684	−1.303	−0.851
45	−3.281	−2.690	−2.412	−2.014	−1.679	−1.301	−0.850
50	−3.261	−2.678	−2.403	−2.009	−1.676	−1.299	−0.849
60	−3.232	−2.660	−2.390	−2.000	−1.671	−1.296	−0.848
70	−3.211	−2.648	−2.381	−1.994	−1.667	−1.294	−0.847
80	−3.195	−2.639	−2.374	−1.990	−1.664	−1.292	−0.846
90	−3.183	−2.632	−2.369	−1.987	−1.662	−1.291	−0.846
100	−3.174	−2.626	−2.364	−1.984	−1.660	−1.290	−0.845
200	−3.131	−2.601	−2.345	−1.972	−1.652	−1.286	−0.843
500	−3.107	−2.586	−2.334	−1.965	−1.648	−1.283	−0.842
1000	−3.098	−2.581	−2.330	−1.962	−1.646	−1.282	−0.842

TABLA F (continuación) Valores de cuantiles de la distribución t de Student

ν	$t_{0.800}$	$t_{0.900}$	$t_{0.950}$	$t_{0.975}$	$t_{0.990}$	$t_{0.995}$	$t_{0.999}$
1	1.376	3.078	6.314	12.706	31.820	63.656	318.294
2	1.061	1.886	2.920	4.303	6.965	9.925	22.327
3	0.978	1.638	2.353	3.182	4.541	5.841	10.214
4	0.941	1.533	2.132	2.776	3.747	4.604	7.173
5	0.920	1.476	2.015	2.571	3.365	4.032	5.893
6	0.906	1.440	1.943	2.447	3.143	3.707	5.208
7	0.896	1.415	1.895	2.365	2.998	3.499	4.785
8	0.889	1.397	1.860	2.306	2.896	3.355	4.501
9	0.883	1.383	1.833	2.262	2.821	3.250	4.297
10	0.879	1.372	1.812	2.228	2.764	3.169	4.144
11	0.876	1.363	1.796	2.201	2.718	3.106	4.025
12	0.873	1.356	1.782	2.179	2.681	3.055	3.930
13	0.870	1.350	1.771	2.160	2.650	3.012	3.852
14	0.868	1.345	1.761	2.145	2.624	2.977	3.787
15	0.866	1.341	1.753	2.131	2.602	2.947	3.733
16	0.865	1.337	1.746	2.120	2.583	2.921	3.686
17	0.863	1.333	1.740	2.110	2.567	2.898	3.646
18	0.862	1.330	1.734	2.101	2.552	2.878	3.610
19	0.861	1.328	1.729	2.093	2.539	2.861	3.579
20	0.860	1.325	1.725	2.086	2.528	2.845	3.552
21	0.859	1.323	1.721	2.080	2.518	2.831	3.527
22	0.858	1.321	1.717	2.074	2.508	2.819	3.505
23	0.858	1.319	1.714	2.069	2.500	2.807	3.485
24	0.857	1.318	1.711	2.064	2.492	2.797	3.467
25	0.856	1.316	1.708	2.060	2.485	2.787	3.450
26	0.856	1.315	1.706	2.056	2.479	2.779	3.435
27	0.855	1.314	1.703	2.052	2.473	2.771	3.421
28	0.855	1.313	1.701	2.048	2.467	2.763	3.408
29	0.854	1.311	1.699	2.045	2.462	2.756	3.396
30	0.854	1.310	1.697	2.042	2.457	2.750	3.385
35	0.852	1.306	1.690	2.030	2.438	2.724	3.340
40	0.851	1.303	1.684	2.021	2.423	2.704	3.307
45	0.850	1.301	1.679	2.014	2.412	2.690	3.281
50	0.849	1.299	1.676	2.009	2.403	2.678	3.261
60	0.848	1.296	1.671	2.000	2.390	2.660	3.232
70	0.847	1.294	1.667	1.994	2.381	2.648	3.211
80	0.846	1.292	1.664	1.990	2.374	2.639	3.195
90	0.846	1.291	1.662	1.987	2.368	2.632	3.183
100	0.845	1.290	1.660	1.984	2.364	2.626	3.174
200	0.843	1.286	1.652	1.972	2.345	2.601	3.131
500	0.842	1.283	1.648	1.965	2.334	2.586	3.107
1000	0.842	1.282	1.646	1.962	2.330	2.581	3.098

TABLA E Valores de cuantiles de la distribución chi-cuadrada

$$P(X \leqslant x_{1-\alpha,\nu}) = F\,(x_{1-\alpha};\,\nu) = \frac{1}{\Gamma(\nu/2)\,2^{\nu/2}} \int_0^{x_{1-\alpha,\nu}} t^{\nu/2-1} \exp(-t/2)\,dt \doteq 1-\alpha$$



$x_{1-\alpha,\nu}$

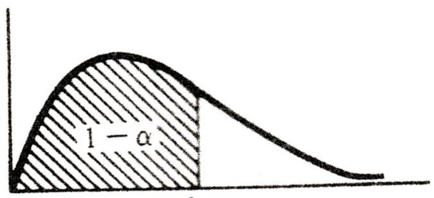
ν	$x_{0.005}$	$x_{0.010}$	$x_{0.025}$	$x_{0.050}$	$x_{0.100}$	$x_{0.900}$	$x_{0.950}$	$x_{0.975}$	$x_{0.990}$	$x_{0.995}$
1	0.00	0.00	0.00	0.00	0.02	2.71	3.84	5.02	6.64	7.90
2	0.01	0.02	0.05	0.10	0.21	4.60	5.99	7.38	9.22	10.59
3	0.07	0.11	0.22	0.35	0.58	6.25	7.82	9.36	11.32	12.82
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.15	13.28	14.82
5	0.41	0.55	0.83	1.15	1.61	9.24	11.07	12.84	15.09	16.76
6	0.67	0.87	1.24	1.63	2.20	10.65	12.60	14.46	16.81	18.55
7	0.99	1.24	1.69	2.17	2.83	12.02	14.07	16.02	18.47	20.27
8	1.34	1.64	2.18	2.73	3.49	13.36	15.51	17.55	20.08	21.94
9	1.73	2.09	2.70	3.32	4.17	14.69	16.93	19.03	21.65	23.56
10	2.15	2.55	3.24	3.94	4.86	15.99	18.31	20.50	23.19	25.15
11	2.60	3.05	3.81	4.57	5.58	17.28	19.68	21.93	24.75	26.71
12	3.06	3.57	4.40	5.22	6.30	18.55	21.03	23.35	26.25	28.25
13	3.56	4.10	5.01	5.89	7.04	19.81	22.37	24.75	27.72	29.88
14	4.07	4.65	5.62	6.57	7.79	21.07	23.69	26.13	29.17	31.38
15	4.59	5.23	6.26	7.26	8.55	22.31	25.00	27.50	30.61	32.86
16	5.14	5.81	6.90	7.96	9.31	23.55	26.30	28.86	32.03	34.32
17	5.69	6.40	7.56	8.67	10.08	24.77	27.59	30.20	33.43	35.77
18	6.25	7.00	8.23	9.39	10.86	25.99	28.88	31.54	34.83	37.21
19	6.82	7.63	8.90	10.11	11.65	27.21	30.15	32.87	36.22	38.63
20	7.42	8.25	9.59	10.85	12.44	28.42	31.42	34.18	37.59	40.05

TABLA E (continuación) Valores de cuantiles de la distribución chi-cuadrada

ν	$x_{0.005}$	$x_{0.010}$	$x_{0.025}$	$x_{0.050}$	$x_{0.100}$	$x_{0.900}$	$x_{0.950}$	$x_{0.975}$	$x_{0.990}$	$x_{0.995}$
21	8.02	8.89	10.28	11.59	13.24	29.62	32.68	35.49	38.96	41.45
22	8.62	9.53	10.98	12.34	14.04	30.82	33.93	36.79	40.31	42.84
23	9.25	10.19	11.69	13.09	14.85	32.01	35.18	38.09	41.66	44.23
24	9.87	10.85	12.40	13.84	15.66	33.20	36.42	39.38	43.00	45.60
25	10.50	11.51	13.11	14.61	16.47	34.38	37.66	40.66	44.34	46.97
26	11.13	12.19	13.84	15.38	17.29	35.57	38.89	41.94	45.66	48.33
27	11.79	12.87	14.57	16.15	18.11	36.74	40.12	43.21	46.99	49.69
28	12.44	13.55	15.30	16.92	18.94	37.92	41.34	44.47	48.30	51.04
29	13.09	14.24	16.04	17.70	19.77	39.09	42.56	45.74	49.61	52.38
30	13.77	14.94	16.78	18.49	20.60	40.26	43.78	46.99	50.91	53.71
35	17.16	18.49	20.56	22.46	24.79	46.06	49.81	53.22	57.36	60.31
40	20.67	22.14	24.42	26.51	29.06	51.80	55.75	59.34	63.71	66.80
45	24.28	25.88	28.36	30.61	33.36	57.50	61.65	65.41	69.98	73.20
50	27.96	29.68	32.35	34.76	37.69	63.16	67.50	71.42	76.17	79.52
60	35.50	37.46	40.47	43.19	46.46	74.39	79.08	83.30	88.40	91.98
70	43.25	45.42	48.75	51.74	55.33	85.52	90.53	95.03	100.44	104.24
80	51.14	53.52	57.15	60.39	64.28	96.57	101.88	106.63	112.34	116.35
90	59.17	61.74	65.64	69.13	73.29	107.56	113.14	118.14	124.13	128.32
100	67.30	70.05	74.22	77.93	82.36	118.49	124.34	129.56	135.82	140.19

TABLA G Valores de cuantiles de la distribución F

$$P(F \leq f_{1-\alpha,\nu_1,\nu_2}) = \frac{\Gamma[(\nu_1+\nu_2)/2]\nu_1^{\nu_1/2}\,\nu_2^{\nu_2/2}}{\Gamma(\nu_1/2)\Gamma(\nu_2/2)} \int_0^{f_{1-\alpha,\nu_1,\nu_2}} t^{(\nu_1-2)/2}\,(\nu_2+\nu_1 t)^{(\nu_1+\nu_2)/2}\,dt = 1-\alpha$$



$f_{1-\alpha,\nu_1,\nu_2}$

$1-\alpha = 0.9$

ν_2	ν_1 = *grados de libertad del numerador* 1	2	3	4	5	6	7	8	9	10
1	39.86	49.50	53.59	55.83	57.24	58.20	58.91	59.44	59.86	60.19
2	8.53	9.00	9.16	9.24	9.29	9.33	9.35	9.37	9.38	9.39
3	5.54	5.46	5.39	5.34	5.31	5.28	5.27	5.25	5.24	5.23
4	4.54	4.32	4.19	4.11	4.05	4.01	3.98	3.95	3.94	3.92
5	4.06	3.78	3.62	3.52	3.45	3.40	3.37	3.34	3.32	3.30
6	3.78	3.46	3.29	3.18	3.11	3.05	3.01	2.98	2.96	2.94
7	3.59	3.26	3.07	2.96	2.88	2.83	2.79	2.75	2.72	2.70
8	3.46	3.11	2.92	2.81	2.73	2.67	2.62	2.59	2.56	2.54
9	3.36	3.01	2.81	2.69	2.61	2.55	2.51	2.47	2.44	2.42
10	3.29	2.92	2.73	2.61	2.52	2.46	2.41	2.38	2.35	2.32
11	3.23	2.86	2.66	2.54	2.45	2.39	2.34	2.30	2.27	2.25
12	3.18	2.81	2.61	2.48	2.39	2.33	2.28	2.24	2.21	2.19
13	3.14	2.76	2.56	2.43	2.35	2.28	2.23	2.20	2.16	2.14
14	3.10	2.73	2.52	2.39	2.31	2.24	2.19	2.15	2.12	2.10
15	3.07	2.70	2.49	2.36	2.27	2.21	2.16	2.12	2.09	2.06
16	3.05	2.67	2.46	2.33	2.24	2.18	2.13	2.09	2.06	2.03
17	3.03	2.64	2.44	2.31	2.22	2.15	2.10	2.06	2.03	2.00
18	3.01	2.62	2.42	2.29	2.20	2.13	2.08	2.04	2.00	1.98
19	2.99	2.61	2.40	2.27	2.18	2.11	2.06	2.02	1.98	1.96
20	2.97	2.59	2.38	2.25	2.16	2.09	2.04	2.00	1.96	1.94
21	2.96	2.57	2.36	2.23	2.14	2.08	2.02	1.98	1.95	1.92
22	2.95	2.56	2.35	2.22	2.13	2.06	2.01	1.97	1.93	1.90
23	2.94	2.55	2.34	2.21	2.11	2.05	1.99	1.95	1.92	1.89
24	2.93	2.54	2.33	2.19	2.10	2.04	1.98	1.94	1.91	1.88
25	2.92	2.53	2.32	2.18	2.09	2.02	1.97	1.93	1.89	1.87
26	2.91	2.52	2.31	2.17	2.08	2.01	1.96	1.92	1.88	1.86
27	2.90	2.51	2.30	2.17	2.07	2.00	1.95	1.91	1.87	1.85
28	2.89	2.50	2.29	2.16	2.06	2.00	1.94	1.90	1.87	1.84
29	2.89	2.50	2.28	2.15	2.06	1.99	1.93	1.89	1.86	1.83
30	2.88	2.49	2.28	2.14	2.05	1.98	1.93	1.88	1.85	1.82
35	2.85	2.46	2.25	2.11	2.02	1.95	1.90	1.85	1.82	1.79
40	2.84	2.44	2.23	2.09	2.00	1.93	1.87	1.83	1.79	1.76
50	2.81	2.41	2.20	2.06	1.97	1.90	1.84	1.80	1.76	1.73
60	2.79	2.39	2.18	2.04	1.95	1.87	1.82	1.77	1.74	1.71
80	2.77	2.37	2.15	2.02	1.92	1.85	1.79	1.75	1.71	1.68
100	2.76	2.36	2.14	2.00	1.91	1.83	1.78	1.73	1.69	1.66
200	2.73	2.33	2.11	1.97	1.88	1.80	1.75	1.70	1.66	1.63
500	2.72	2.31	2.09	1.96	1.86	1.79	1.73	1.68	1.64	1.61
1000	2.71	2.31	2.09	1.95	1.85	1.78	1.72	1.68	1.64	1.61

TABLA G (continuación) Valores de cuantiles de la distribución F

$1 - \alpha = 0.9$

	ν_1 = *grados de libertad del numerador*									
ν_2	11	12	15	20	25	30	40	50	100	1000
1	60.47	60.71	61.22	61.74	62.06	62.26	62.53	62.69	63.00	63.29
2	9.40	9.41	9.42	9.44	9.45	9.46	9.47	9.47	9.48	9.49
3	5.22	5.22	5.20	5.19	5.17	5.17	5.16	5.15	5.14	5.13
4	3.91	3.90	3.87	3.84	3.83	3.82	3.80	3.80	3.78	3.76
5	3.28	3.27	3.24	3.21	3.19	3.17	3.16	3.15	3.13	3.11
6	2.92	2.90	2.87	2.84	2.81	2.80	2.78	2.77	2.75	2.72
7	2.68	2.67	2.63	2.59	2.57	2.56	2.54	2.52	2.50	2.47
8	2.52	2.50	2.46	2.42	2.40	2.38	2.36	2.35	2.32	2.30
9	2.40	2.38	2.34	2.30	2.27	2.25	2.23	2.22	2.19	2.16
10	2.30	2.28	2.24	2.20	2.17	2.16	2.13	2.12	2.09	2.06
11	2.23	2.21	2.17	2.12	2.10	2.08	2.05	2.04	2.00	1.98
12	2.17	2.15	2.10	2.06	2.03	2.01	1.99	1.97	1.94	1.91
13	2.12	2.10	2.05	2.01	1.98	1.96	1.93	1.92	1.88	1.85
14	2.07	2.05	2.01	1.96	1.93	1.91	1.89	1.87	1.83	1.80
15	2.04	2.02	1.97	1.92	1.89	1.87	1.85	1.83	1.79	1.76
16	2.01	1.99	1.94	1.89	1.86	1.84	1.81	1.79	1.76	1.72
17	1.98	1.96	1.91	1.86	1.83	1.81	1.78	1.76	1.73	1.69
18	1.95	1.93	1.89	1.84	1.80	1.78	1.75	1.74	1.70	1.66
19	1.93	1.91	1.86	1.81	1.78	1.76	1.73	1.71	1.67	1.64
20	1.91	1.89	1.84	1.79	1.76	1.74	1.71	1.69	1.65	1.61
21	1.90	1.87	1.83	1.78	1.74	1.72	1.69	1.67	1.63	1.59
22	1.88	1.86	1.81	1.76	1.73	1.70	1.67	1.65	1.61	1.57
23	1.87	1.84	1.80	1.74	1.71	1.69	1.66	1.64	1.59	1.55
24	1.85	1.83	1.78	1.73	1.70	1.67	1.64	1.62	1.58	1.54
25	1.84	1.82	1.77	1.72	1.68	1.66	1.63	1.61	1.56	1.52
26	1.83	1.81	1.76	1.71	1.67	1.65	1.61	1.59	1.55	1.51
27	1.82	1.80	1.75	1.70	1.66	1.64	1.60	1.58	1.54	1.50
28	1.81	1.79	1.74	1.69	1.65	1.63	1.59	1.57	1.53	1.48
29	1.80	1.78	1.73	1.68	1.64	1.62	1.58	1.56	1.52	1.47
30	1.79	1.77	1.72	1.67	1.63	1.61	1.57	1.55	1.51	1.46
35	1.76	1.74	1.69	1.63	1.60	1.57	1.53	1.51	1.47	1.42
40	1.74	1.71	1.66	1.61	1.57	1.54	1.51	1.48	1.43	1.38
50	1.70	1.68	1.63	1.57	1.53	1.50	1.46	1.44	1.39	1.33
60	1.68	1.66	1.60	1.54	1.50	1.48	1.44	1.41	1.36	1.30
80	1.65	1.63	1.57	1.51	1.47	1.44	1.40	1.38	1.32	1.25
100	1.64	1.61	1.56	1.49	1.45	1.42	1.38	1.35	1.29	1.22
200	1.60	1.58	1.52	1.46	1.41	1.38	1.34	1.31	1.24	1.16
500	1.58	1.56	1.50	1.44	1.39	1.36	1.31	1.28	1.21	1.11
1000	1.58	1.55	1.49	1.43	1.38	1.35	1.30	1.27	1.20	1.08

TABLA G (continuación) Valores de cuantiles de la distribución F

$1 - \alpha = 0.95$

	ν_1 = *grados de libertad del numerador*									
ν_2	1	2	3	4	5	6	7	8	9	10
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.97
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.73
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16
35	4.12	3.27	2.87	2.64	2.49	2.37	2.29	2.22	2.16	2.11
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08
50	4.03	3.18	2.79	2.56	2.40	2.29	2.20	2.13	2.07	2.03
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99
80	3.96	3.11	2.72	2.49	2.33	2.21	2.13	2.06	2.00	1.95
100	3.94	3.09	2.70	2.46	2.31	2.19	2.10	2.03	1.97	1.93
200	3.89	3.04	2.65	2.42	2.26	2.14	2.06	1.98	1.93	1.88
500	3.86	3.01	2.62	2.39	2.23	2.12	2.03	1.96	1.90	1.85
1000	3.85	3.01	2.61	2.38	2.22	2.11	2.02	1.95	1.89	1.84

TABLA G (continuación) Valores de cuantiles de la distribución F

$1 - \alpha = 0.95$

	ν_1 = *grados de libertad del numerador*									
ν_2	11	12	15	20	25	30	40	50	100	1000
1	242.98	243.91	245.96	248.01	249.26	250.08	251.15	251.77	253.01	254.17
2	19.40	19.41	19.43	19.45	19.46	19.46	19.47	19.48	19.49	19.50
3	8.76	8.74	8.70	8.66	8.63	8.62	8.59	8.58	8.55	8.53
4	5.94	5.91	5.86	5.80	5.77	5.74	5.72	5.70	5.66	5.63
5	4.70	4.68	4.62	4.56	4.52	4.50	4.46	4.44	4.41	4.37
6	4.03	4.00	3.94	3.87	3.84	3.81	3.77	3.75	3.71	3.67
7	3.60	3.57	3.51	3.44	3.40	3.38	3.34	3.32	3.27	3.23
8	3.31	3.28	3.22	3.15	3.11	3.08	3.04	3.02	2.97	2.93
9	3.10	3.07	3.01	2.94	2.89	2.86	2.83	2.80	2.76	2.71
10	2.94	2.91	2.85	2.77	2.73	2.70	2.66	2.64	2.59	2.54
11	2.82	2.79	2.72	2.65	2.60	2.57	2.53	2.51	2.46	2.41
12	2.72	2.69	2.62	2.54	2.50	2.47	2.43	2.40	2.35	2.30
13	2.63	2.60	2.53	2.46	2.41	2.38	2.34	2.31	2.26	2.21
14	2.57	2.53	2.46	2.39	2.34	2.31	2.27	2.24	2.19	2.14
15	2.51	2.48	2.40	2.33	2.28	2.25	2.20	2.18	2.12	2.07
16	2.46	2.42	2.35	2.28	2.23	2.19	2.15	2.12	2.07	2.02
17	2.41	2.38	2.31	2.23	2.18	2.15	2.10	2.08	2.02	1.97
18	2.37	2.34	2.27	2.19	2.14	2.11	2.06	2.04	1.98	1.92
19	2.34	2.31	2.23	2.16	2.11	2.07	2.03	2.00	1.94	1.88
20	2.31	2.28	2.20	2.12	2.07	2.04	1.99	1.97	1.91	1.85
21	2.28	2.25	2.18	2.10	2.05	2.01	1.96	1.94	1.88	1.82
22	2.26	2.23	2.15	2.07	2.02	1.98	1.94	1.91	1.85	1.79
23	2.24	2.20	2.13	2.05	2.00	1.96	1.91	1.88	1.82	1.76
24	2.22	2.18	2.11	2.03	1.97	1.94	1.89	1.86	1.80	1.74
25	2.20	2.16	2.09	2.01	1.96	1.92	1.87	1.84	1.78	1.72
26	2.18	2.15	2.07	1.99	1.94	1.90	1.85	1.82	1.76	1.70
27	2.17	2.13	2.06	1.97	1.92	1.88	1.84	1.81	1.74	1.68
28	2.15	2.12	2.04	1.96	1.91	1.87	1.82	1.79	1.73	1.66
29	2.14	2.10	2.03	1.94	1.89	1.85	1.81	1.77	1.71	1.65
30	2.13	2.09	2.01	1.93	1.88	1.84	1.79	1.76	1.70	1.63
35	2.07	2.04	1.96	1.88	1.82	1.79	1.74	1.70	1.63	1.57
40	2.04	2.00	1.92	1.84	1.78	1.74	1.69	1.66	1.59	1.52
50	1.99	1.95	1.87	1.78	1.73	1.69	1.63	1.60	1.52	1.45
60	1.95	1.92	1.84	1.75	1.69	1.65	1.59	1.56	1.48	1.40
80	1.91	1.88	1.79	1.70	1.64	1.60	1.54	1.51	1.43	1.34
100	1.89	1.85	1.77	1.68	1.62	1.57	1.52	1.48	1.39	1.30
200	1.84	1.80	1.72	1.62	1.56	1.52	1.46	1.41	1.32	1.21
500	1.81	1.77	1.69	1.59	1.53	1.48	1.42	1.38	1.28	1.14
1000	1.80	1.76	1.68	1.58	1.52	1.47	1.41	1.36	1.26	1.11

TABLA G (continuación) Valores de cuantiles de la distribución *F*

$1 - \alpha = 0.99$

ν_2	ν_1 = *grados de libertad del numerador* 1	2	3	4	5	6	7	8	9	10
2	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39	99.40
3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.50	27.34	27.22
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10
14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03	3.94
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69
17	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68	3.59
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51
19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40	3.31
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26
23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17
25	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.32	3.22	3.13
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09
27	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26	3.15	3.06
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03
29	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.20	3.09	3.00
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98
35	7.42	5.27	4.40	3.91	3.59	3.37	3.20	3.07	2.96	2.88
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80
50	7.17	5.06	4.20	3.72	3.41	3.19	3.02	2.89	2.78	2.70
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63
80	6.96	4.88	4.04	3.56	3.26	3.04	2.87	2.74	2.64	2.55
100	6.90	4.82	3.98	3.51	3.21	2.99	2.82	2.69	2.59	2.50
200	6.76	4.71	3.88	3.41	3.11	2.89	2.73	2.60	2.50	2.41
500	6.69	4.65	3.82	3.36	3.05	2.84	2.68	2.55	2.44	2.36
1000	6.66	4.63	3.80	3.34	3.04	2.82	2.66	2.53	2.43	2.34

TABLA G (continuación) Valores de cuantiles de la distribución F

$1 - \alpha = 0.99$

ν_2	ν_1 = *grados de libertad del numerador* 11	12	15	20	25	30	40	50	100	1000
2	99.41	99.42	99.43	99.45	99.46	99.46	99.47	99.48	99.49	99.51
3	27.12	27.03	26.85	26.67	26.58	26.50	26.41	26.35	26.24	26.14
4	14.45	14.37	14.19	14.02	13.91	13.84	13.75	13.69	13.58	13.48
5	9.96	9.89	9.72	9.55	9.45	9.38	9.30	9.24	9.13	9.03
6	7.79	7.72	7.56	7.40	7.29	7.23	7.15	7.09	6.99	6.89
7	6.54	6.47	6.31	6.16	6.06	5.99	5.91	5.86	5.75	5.66
8	5.73	5.67	5.52	5.36	5.26	5.20	5.12	5.07	4.96	4.87
9	5.18	5.11	4.96	4.81	4.71	4.65	4.57	4.52	4.41	4.32
10	4.77	4.71	4.56	4.41	4.31	4.25	4.17	4.12	4.01	3.92
11	4.46	4.40	4.25	4.10	4.00	3.94	3.86	3.81	3.71	3.61
12	4.22	4.16	4.01	3.86	3.76	3.70	3.62	3.57	3.47	3.37
13	4.02	3.96	3.82	3.66	3.57	3.51	3.43	3.38	3.27	3.18
14	3.86	3.80	3.66	3.51	3.41	3.35	3.27	3.22	3.11	3.02
15	3.73	3.67	3.52	3.37	3.28	3.21	3.13	3.08	2.98	2.88
16	3.62	3.55	3.41	3.26	3.16	3.10	3.02	2.97	2.86	2.76
17	3.52	3.46	3.31	3.16	3.07	3.00	2.92	2.87	2.76	2.66
18	3.43	3.37	3.23	3.08	2.98	2.92	2.84	2.78	2.68	2.58
19	3.36	3.30	3.15	3.00	2.91	2.84	2.76	2.71	2.60	2.50
20	3.29	3.23	3.09	2.94	2.84	2.78	2.69	2.64	2.54	2.43
21	3.24	3.17	3.03	2.88	2.78	2.72	2.64	2.58	2.48	2.37
22	3.18	3.12	2.98	2.83	2.73	2.67	2.58	2.53	2.42	2.32
23	3.14	3.07	2.93	2.78	2.69	2.62	2.54	2.48	2.37	2.27
24	3.09	3.03	2.89	2.74	2.64	2.58	2.49	2.44	2.33	2.22
25	3.06	2.99	2.85	2.70	2.60	2.54	2.45	2.40	2.29	2.18
26	3.02	2.96	2.81	2.66	2.57	2.50	2.42	2.36	2.25	2.14
27	2.99	2.93	2.78	2.63	2.54	2.47	2.38	2.33	2.22	2.11
28	2.96	2.90	2.75	2.60	2.51	2.44	2.35	2.30	2.19	2.08
29	2.93	2.87	2.73	2.57	2.48	2.41	2.33	2.27	2.16	2.05
30	2.91	2.84	2.70	2.55	2.45	2.39	2.30	2.24	2.13	2.02
35	2.80	2.74	2.60	2.44	2.35	2.28	2.19	2.14	2.02	1.90
40	2.73	2.66	2.52	2.37	2.27	2.20	2.11	2.06	1.94	1.82
50	2.62	2.56	2.42	2.27	2.17	2.10	2.01	1.95	1.82	1.70
60	2.56	2.50	2.35	2.20	2.10	2.03	1.94	1.88	1.75	1.62
80	2.48	2.42	2.27	2.12	2.01	1.94	1.85	1.79	1.65	1.51
100	2.43	2.37	2.22	2.07	1.97	1.89	1.80	1.74	1.60	1.45
200	2.34	2.27	2.13	1.97	1.87	1.79	1.69	1.63	1.48	1.30
500	2.28	2.22	2.07	1.92	1.81	1.74	1.63	1.57	1.41	1.20
1000	2.27	2.20	2.06	1.90	1.79	1.72	1.61	1.54	1.38	1.16